

Sigma 6.0 (DSA)

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Automatic Zoom

Graph-02 Solutions

Solution 1:

Time Complexity : $O(V+E)$ Space Complexity: $O(V+E)$

```
import java.util.*;
```

```
class Solution{
```

```
    static void addEdge(int u, int v,  
        ArrayList<ArrayList<Integer>> adj){  
        adj.get(u).add(v);  
    }
```

```
    static void DFSUtil(ArrayList<ArrayList<Integer>> g,  
        int v, boolean[] visited){  
        visited[v] = true;
```

```
        for(int x : g.get(v)){  
            if (!visited[x]){  
                DFSUtil(g, x, visited);  
            }  
        }  
    }
```

```
    static int motherVertex(ArrayList<ArrayList<Integer>>g,  
        int V){
```

```
        boolean[] visited = new boolean[V];  
        int v = -1;
```

```
        for(int i = 0; i < V; i++){  
            if (!visited[i]){  
                DFSUtil(g, i, visited);  
                v = i;  
            }  
        }
```